

Multi-agent BOM procurement *automation.*

An 8-week pilot to automate the ingestion, extraction and reconciliation of engineering BOMs into the SAP Material Master — without compromising data integrity.

PREPARED FOR

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Three numbers *define* the pilot.

CYCLE TIME TODAY

24–48_h

Time-zone gap, HQ → Asia. Dead time.

MANUAL WORKFORCE

15_{FTE}

Triaging an inbox, copy-pasting into SAP.

COST OF ONE ERROR

€350_K

500 t of unusable steel + line delays.

Four handoffs. *Zero* guardrails.

01

Monitor & triage
new / revision / dup

MANUAL

! 24-48H DELAY

02

Extract specs
from PDF / Excel / body

MANUAL

! 3 FORMATS, 0 SCHEMA

03

Reconcile to SAP code
"Galv Steel B" → AC-GALV-TB-001

TRIBAL

! SAP REJECTS LITERAL
TEXT

04

Key PO into SAP

MANUAL

! €350K PER MISREAD

Four agents. *One* human gate.

A1

Extractor

Typed BOM lines from PDF / Excel / body.

AUTO

A2

Validator loop

Self-critique up to 3x.

LOOP

A3

Reconciler

Match to SAP MM with confidence.

AUTO

G

HITL gate

Block PO if any line uncertain.

HUMAN

A4

PO drafter

DRAFT only into SAP sandbox.

AUTO

● AUTOMATED

- Inbox monitoring & dedup
- BOM extraction across formats
- Reconciliation to SAP MM

● HUMAN-IN-THE-LOOP

- Low-confidence reconciliations
- Margin notes that change spec
- One-click PO confirmation

○ OUT OF SCOPE · PILOT

- ~~Live SAP commit~~
- ~~CAD drawing OCR~~
- ~~Non-steel commodities~~

Typed BOM in. *Confident* SAP code out.

```
# Per BOM line · Pydantic
class LineItem:
    line_ref      : str
    raw_description : str
    grade         : str
    form          : Form | None      # sheet/plate/coil/tube...
    thickness_mm   : float | None
    quantity      : float
    uom           : UoM | None      # TON/KG/M
    notes         : str            # margin notes

# Extraction wraps lines with email_type
# (NEW|REVISION|DUP), project, bom_revision.
# ReconciledLine adds sap_code, confidence,
# status (auto/hitl), hitl_reason.
```

FRICTION #2 – TEXT → SAP CODE · LEARNS OVER TIME

1

Ground-truth map · built from history

60 days of past email → PO pairs gives us thousands of "Galv Steel B
→ AC-GALV-TB-001" mappings on day one.

2

Reconciler agent reasons · for new text

Reads candidate SAP codes, applies steel-domain rules (grade family, thickness, tolerance), scores its own confidence.

!

Below threshold → HITL

PO blocked until queue empty. By design.

↺

Every HITL decision feeds the map

The agent learns. Same uncertainty doesn't appear twice — accuracy compounds week over week.

What we need *by* Friday W1.



Read access · shared procurement inbox

A read-only mailbox account so the agent can ingest emails and attachments without sending.

IT DIRECTOR



Read-only API to SAP Material Master

OData or RFC endpoint so the agent can look up valid steel codes live. Read-only — no writes to production.

SAP BASIS TEAM



SAP sandbox + DRAFT-PO service user

A non-production SAP client where the agent can create draft POs for human review.

SAP BASIS TEAM



PHARO console for ops; SAP for audit

Operational telemetry — agent accuracy, queue depth, cost per run — lives in the PHARO console. Approvals and the legal audit trail stay inside SAP. No parallel auth system.

BY DESIGN



Two named experts · 4h / week each

One procurement analyst + one engineering or quality lead.

CPO



Labelled corpus · past 60 days of email → PO pairs

Past emails joined to the SAP POs they produced. This is our benchmark for measuring agent accuracy.

SAP BASIS + IT DIRECTOR

Eight weeks. *Three* phases.

W1 – W2 · 2 WEEKS

Discovery

Extractor proven on the labelled corpus.

+ GATE 1

Extraction accuracy ≥ 95%

W3 – W4 · 2 WEEKS

Build

End-to-end pipeline writing DRAFT POs into the SAP sandbox.

+ GATE 2

≥ 90% auto-drafted correctly

W5 – W8 · 4 WEEKS

Pilot

Agent live on real emails. Performance measured daily.

■ GO-LIVE

≥ 90% auto-approved · < 30 min cycle time

Project plan.

COMMUNICATION

DAILY

15-min check-in with the procurement lead — review the human review queue and any blockers.

WEEKLY

Sponsor review with the CPO + IT Director — progress, risks, scope changes.

BI-WEEKLY

SAP integration review with the IT team — confirm the agent and SAP are still talking cleanly.

ON INCIDENT

Critical alert within 60 min · the agent can be paused in under 5 min.

SUCCESS METRICS · TRACKED LIVE IN *PHARO*

TIME · EMAIL → DRAFT PO

< 30m

TODAY: 24-48H

POS AUTO-APPROVED BY W8

≥ 90%

NO HUMAN REVIEW

WRONG SAP CODES SHIPPED

0

ZERO TOLERANCE

PROCUREMENT HOURS FREED

~60%

OF 15 FTE